

Thermodynamic and Kinematic Analysis of the Stirling Engine

Abstract

The Stirling engine is an external combustion heat engine that operates on a closed, regenerative thermodynamic cycle. By cyclically compressing and expanding a fixed mass of gaseous working fluid (typically air, hydrogen, or helium) at different temperature levels, it achieves a net conversion of thermal energy into mechanical work. This document provides a rigorous engineering analysis of its primary components, ideal thermodynamic phases, energy conversion mechanics, and advanced analytical models such as the Schmidt analysis.

1. Architectural Components and Kinematics

The physical architecture of a Stirling engine consists of several coupled thermodynamic control volumes and kinematic linkages. Configurations are generally classified into three types: **Alpha** (two independent power pistons), **Beta** (displacer and power piston in the same cylinder), and **Gamma** (displacer and power piston in separate cylinders).

1.1 The Expansion Space (Hot Chamber)

The hot chamber represents the high-temperature reservoir (T_H) control volume, denoted as V_E . External heat transfer Q_{in} occurs across the cylinder walls. The physical expansion of the high-temperature, high-pressure gas in this region performs positive boundary work.

1.2 The Compression Space (Cold Chamber)

The cold chamber represents the low-temperature sink (T_C) control volume, denoted as V_C . Heat rejection Q_{out} occurs here, lowering the internal energy of the fluid. This minimizes the required work input during the subsequent compression phase.

1.3 The Displacer (Beta & Gamma Configurations)

Unlike a traditional piston, the displacer does not extract work; it maintains a negligible pressure differential across its body ($\Delta P \approx 0$). Its sole function is fluid shuttling. By moving longitudinally, it sweeps the working fluid through the heat exchangers and the regenerator between V_E and V_C . Kinematically, the displacer's motion typically leads the power piston by a phase angle of approximately $\phi \approx 90^\circ$.

1.4 The Power Piston

The power piston seals the working fluid within the engine and acts as the primary mechanical interface. It is subject to the instantaneous uniform pressure $P(t)$ of the system. The net cycle work is the cyclic integral of the pressure over the swept volume:

$$W_{net} = \oint P dV$$

1.5 Kinematic Energy Storage (Flywheel)

Due to the intermittent nature of power generation in a single-cylinder equivalent cycle, a flywheel provides necessary rotational inertia (I). It acts as a mechanical low-pass filter, storing kinetic energy $E_k = \frac{1}{2}I\omega^2$ during the expansion phase and releasing it to drive the system through the compression and isochoric displacement phases, maintaining a continuous angular velocity ω .

2. Ideal Thermodynamic Cycle (The Four Phases)

The ideal Stirling cycle is modeled using two isothermal processes bounded by two isochoric (constant volume) regenerative processes. Assuming an ideal gas ($PV = mRT$), the states transition as follows:

Phase 1: Isochoric Heating (State 4 $\rightarrow$ State 1)

- **Physical Action:** The fluid is moved from the cold chamber to the hot chamber. The total volume remains constant ($V_1 = V_4$).
- **Thermodynamics:** The working fluid passes through the regenerator and hot heat exchanger, absorbing thermal energy. The temperature rises from T_C to T_H , and the pressure increases proportionally ($P_1 = P_4 \frac{T_H}{T_C}$). No mechanical work is done ($W_{4-1} = 0$). Entropy increases as $ds = c_v \frac{dT}{T}$.

Phase 2: Isothermal Expansion (State 1 $\rightarrow$ State 2)

- **Physical Action:** The high pressure drives the power piston outward. The volume increases from V_1 to V_2 .
- **Thermodynamics:** Heat (Q_{in}) is continuously supplied from the external source to maintain a constant maximum temperature (T_H). The fluid performs maximum positive boundary work:

$$W_{1-2} = Q_{in} = \int_{V_1}^{V_2} P dV = mRT_H \ln \left(\frac{V_2}{V_1} \right)$$

Phase 3: Isochoric Cooling (State 2 $\rightarrow$ State 3)

- **Physical Action:** The fluid is moved back to the cold chamber. The volume remains at maximum ($V_2 = V_3$).
- **Thermodynamics:** The fluid transfers thermal energy to the regenerator (storing it for Phase 1). The temperature drops from T_H to T_C , causing a dramatic drop in pressure. No work is done ($W_{2-3} = 0$).

Phase 4: Isothermal Compression (State 3 → State 4)

- **Physical Action:** The flywheel's momentum drives the power piston inward, compressing the fluid from V_3 back to V_4 .
- **Thermodynamics:** The fluid is maintained at T_C by rejecting heat (Q_{out}) to the environment. Compression work is minimized:

$$W_{3-4} = Q_{out} = \int_{V_3}^{V_4} P dV = mRT_C \ln \left(\frac{V_4}{V_3} \right)$$

3. Energy Conversion and System Efficiency

The net mechanical work output per cycle is the difference between expansion and compression work:

$$W_{net} = W_{1-2} + W_{3-4} = mR \ln(r)(T_H - T_C)$$

Where $r = V_2/V_1$ is the volumetric compression ratio.

Thermal Efficiency (η_{th})

Assuming 100% regenerator effectiveness ($\epsilon = 1$), the thermal efficiency perfectly matches the theoretical Carnot limit. It depends solely on the absolute temperatures of the hot and cold reservoirs:

$$\eta_{th} = \frac{W_{net}}{Q_{in}} = 1 - \frac{T_C}{T_H}$$

4. Advanced Kinematics: Schmidt Isothermal Analysis

In a practical engine, pistons undergo harmonic (sinusoidal) motion, meaning isochoric and isothermal phases overlap. Gustav Schmidt (1871) developed a classical analysis based on several assumptions: isothermal working spaces, perfect regeneration, and ideal gas behavior.

For a sinusoidal drive mechanism, the instantaneous volumes of the expansion (V_E) and compression (V_C) spaces as a function of the crank angle $\theta = \omega t$ are:

$$V_E(\theta) = \frac{V_{SE}}{2}(1 - \cos \theta)$$

$$V_C(\theta) = \frac{V_{SC}}{2}[1 - \cos(\theta - \alpha)]$$

Where V_{SE} and V_{SC} are the swept volumes, and α is the phase angle by which the expansion space leads the compression space.

Because the total mass of the working fluid m is constant, we apply the ideal gas law across all internal volumes (including the constant dead volume of the regenerator V_R at effective temperature T_R):

$$m = \frac{P(\theta)}{R} \left(\frac{V_E(\theta)}{T_H} + \frac{V_R}{T_R} + \frac{V_C(\theta)}{T_C} \right)$$

This allows us to solve for the instantaneous pressure $P(\theta)$ as a function of crank angle. The integration of $P(\theta) \frac{dV_T}{d\theta}$ over one full cycle yields the Schmidt work output, which accurately predicts the elliptical shape of a real-world $P - V$ diagram.

5. The Regenerator Matrix Thermodynamics

The regenerator is the critical thermal capacitor of the Stirling engine. It is typically constructed from tightly packed wire mesh or metallic foils. Its function is to absorb heat from the gas during the isochoric cooling phase and return it during the isochoric heating phase.

The thermal effectiveness (ϵ) of the regenerator drastically dictates actual engine efficiency. It is defined as the ratio of actual enthalpy transfer to the theoretical maximum:

$$\epsilon = \frac{T_{fluid,out} - T_{fluid,in}}{T_{matrix,max} - T_{matrix,min}}$$

If $\epsilon < 1$, the external heat source must supply additional thermal energy to bridge the gap, which severely degrades overall thermal efficiency below the Carnot limit. However, increasing regenerator density to maximize ϵ simultaneously increases aerodynamic flow friction (ΔP_{drop}) and dead volume (V_R), creating a complex multidisciplinary optimization problem.

6. Working Fluid Thermophysics

The choice of working fluid critically impacts the engine's power density and specific output.

- **Air:** Convenient and safe, but possesses low thermal conductivity (k) and a high molar mass, limiting the engine to low operational speeds due to severe flow losses.
- **Hydrogen (H_2):** The thermodynamically ideal fluid. It possesses the lowest viscosity (μ) and highest thermal conductivity, leading to exceptionally high heat transfer rates with minimal aerodynamic pumping losses. However, its high diffusivity causes containment issues (embrittlement and leakage), and it poses significant flammability risks.
- **Helium (He):** The standard for industrial and space-rated Stirling engines (e.g., NASA's Advanced Stirling Radioisotope Generator). As a noble gas, it is inert and safe. While its thermal conductivity is slightly lower than hydrogen, its dynamic viscosity is comparable, and being a monatomic gas, its specific heat ratio ($\gamma = c_p/c_v \approx 1.66$) leads to substantial